

The Grand Marina Hotel – HYDROLOGIC Water Monitoring System

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1. EXECUTIVE SUMMARY

Security testing revealed that the HYDROLOGIC system is currently vulnerable to "replay attacks," where an intruder captures and re-sends old sensor data to trick the hotel's dashboard. We tested three individual defense layers and determined that only a combined approach provides 100% protection against all attack types. We recommend immediate implementation of this "triple-layer" defense to prevent costly false alarms and undetected water damage.

2. THE PROBLEM

Replay attacks, they capture data logs (sensor readings). This involves modifying the readings and resending previous readings at different times.

What could go wrong:

- **Modified:** Attackers can modify a sensor reading to seem like a pipe is leaking when it's not leading to the sensor being turned off
- **Resending:** Attackers can resend a sensor reading which makes it seem like there are no water leaks when there can be.

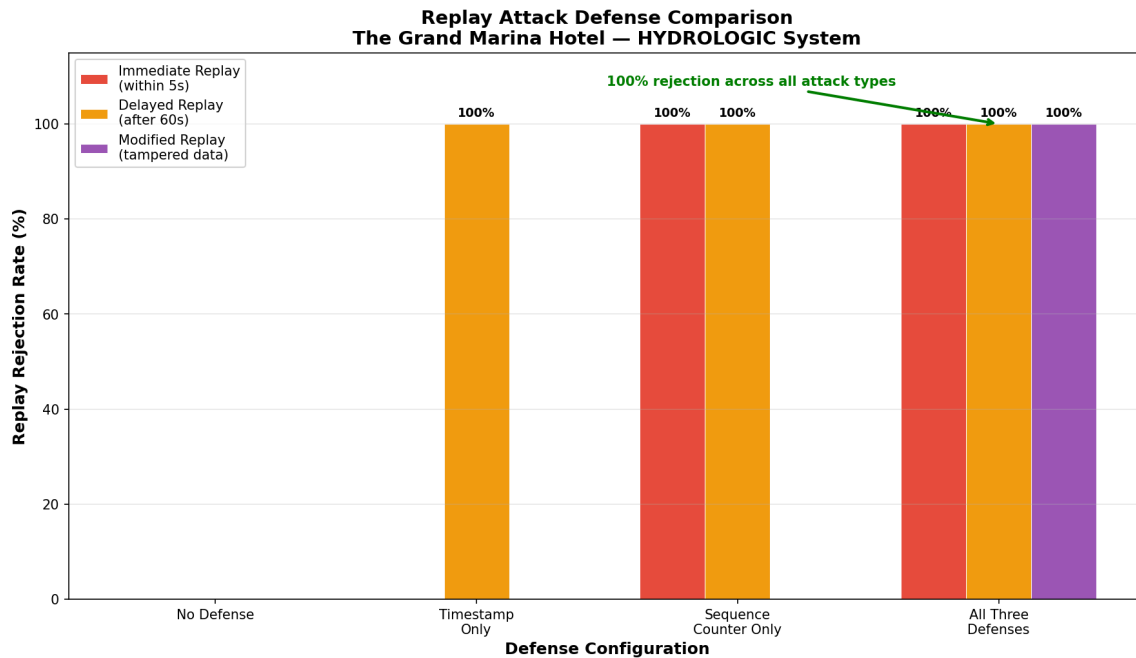
3. DEFENSES TESTED

- **Message freshness check** – Each reading is given an expiration window, anything outside that window will be rejected, inside will be verified as a legitimate sensor reading.
- **Message numbering** – Each reading gets a unique number that verifies it's identity, this gives defense to delayed attacks or immediate.
- **Message signing** – Each reading is verified by a signature which is great for tamper proof if the attacker was to modify the reading for a potential false leakage alert.

4. TESTING RESULTS

Protection Setup	Immediate Replay	Delayed Replay	Tampered Replay
No protection	[0 %]	[0 %]	[0 %]
[Protection 1] only	[0 %]	[100 %]	[_ %]

[Protection 2] only	[100 %]	[100 %]	[0 %]
All three combined	[100 %]	[100 %]	[100 %]



Key finding: Table shows the rejection rate of each defense and attack, No defense has a 0% rejection all around, Timestamp Only has 100% rejection on only attacks after the given expiration window, Sequence Counter Only has 100% on both immediate attacks and delayed but 0% on modified, All defenses grant 100% security all across as it implements a signature.

5. RECOMMENDATION

I recommend All defenses become implemented for full protection against different replay attacks.

- **Modified attacks:** If modified attacks occur there is a risk of false readings being sent to the engineers dashboard which can lead to a sensor being shut off causing thousands of dollars depending on the damages.
- **Resending logs:** If a critical alert for a water leakage was sent 2 hours ago and is replayed after a fix, this can cost a lot as well due to sensors being shut off again and customers not gaining access to water.

6. NEXT STEPS

- **Approval for Implementation:** Authorize the IT and Engineering teams to push the updated security firmware to all guest floor sensors.
- **Update Insurance Profile:** Provide these test results to Sandra Chen to confirm the hotel has met the "Cyber-Physical Security" requirements for the current policy year.
- **Staff Briefing:** Briefly train the maintenance staff on how the new "Security Alert" notifications will appear on their mobile tablets if a tampered message is detected.

7. INDUSTRY REFERENCES

- **AWS IoT Core:** Uses "Message Deduplication" logic, which mirrors our **Message Numbering** to ensure no two messages are processed twice.
- **IEC 62443:** This international industrial standard requires "Integrity and Authenticity" checks, which we have fulfilled through our **Message Signing** protocol.
- **MQTT v5 Protocol:** The latest standard for smart-building communication includes a "Message Expiry Interval," validating our use of **Message Freshness** to discard stale data.